

3.1 Warm-Up Exercises

3.1. Determine parametric equations for the line ℓ in the following cases:

- a) ℓ contains the point $A(1, 2)$ and is parallel to the vector $\mathbf{a}(3, -1)$,
- b) ℓ contains the origin and is parallel to $\mathbf{b}(4, 5)$,
- c) ℓ contains the point $M(1, 7)$ and is parallel to Oy ,
- d) ℓ contains the points $M(2, 4)$ and $N(2, -5)$.

3.2. For the lines ℓ in the previous exercise

- a) determine a Cartesian equation for ℓ ,
- b) describe all direction vectors for ℓ .

3.3. Determine parametric equations for the plane π in the following cases:

- a) π contains the point $M(1, 0, 2)$ and is parallel to the vectors $\mathbf{a}_1(3, -1, 1)$ and $\mathbf{a}_2(0, 3, 1)$,
- b) π contains the points $A(-2, 1, 1)$, $B(0, 2, 3)$ and $C(1, 0, -1)$,
- c) π contains the point $A(1, 2, 1)$ and is parallel to $\mathbf{i}$ and $\mathbf{j}$,
- d) π contains the point $M(1, 7, 1)$ and is parallel to the coordinate plane Oyz ,
- e) π contains the points $M_1(5, 3, 4)$, $M_2(1, 0, 1)$ and is parallel to the vector $\mathbf{a}(1, 3, -3)$,
- f) π contains the point $A(1, 5, 7)$ and the coordinate axis Ox .

3.4. Determine Cartesian equations for the plane π in the following cases:

- a) $\pi : x = 2 + 3u - 4v, y = 4 - v, z = 2 + 3u$ with $u, v \in \mathbb{R}$,
- b) $\pi : x = u + v, y = u - v, z = 5 + 6u - 4v$ with $u, v \in \mathbb{R}$.

3.5. Determine parametric equations for the plane π in the following cases:

- a) $\pi : 3x - 6y + z = 0$,
- b) $\pi : 2x - y - z - 3 = 0$.

3.6. Show that the points $A(1, 0, -1)$, $B(0, 2, 3)$, $C(-2, 1, 1)$ and $D(4, 2, 3)$ are coplanar.

3.7. Determine parametric equations for the line ℓ in the following cases:

- a) ℓ contains the point $M_0(2, 0, 3)$ and is parallel to the vector $\mathbf{a}(3, -2, -2)$,
- b) ℓ contains the point $A(1, 2, 3)$ and is parallel to the Oz -axis,
- c) ℓ contains the points $M_1(1, 2, 3)$ and $M_2(4, 4, 4)$.

3.8. Give Cartesian equations for the lines ℓ in the previous exercise.

3.9. Determine parametric equations for the line contained in the planes $x + y + 2z - 3 = 0$ and $x - y + z - 1 = 0$.

3.10. In each of the following, find a Cartesian equation for the plane containing the point Q and the line ℓ .

a) $Q = (3, 3, 1)$, $\ell : x = 2 + 3t, y = 5 + t, z = 1 + 7t$ with $t \in \mathbb{R}$,

b) $Q = (2, 1, 0)$, $\ell : x - y + 1 = 0, 3x + 5z - 7 = 0$.

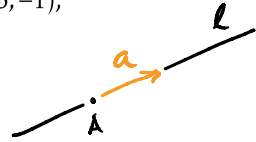
Solutions



3.1. Determine parametric equations for the line ℓ in the following cases:

a) ℓ contains the point $A(1, 2)$ and is parallel to the vector $\mathbf{a}(3, -1)$,

$$\ell: \begin{cases} x = 1 + 3t \\ y = 2 - t \end{cases} \quad t \in \mathbb{R}$$



$$\Leftrightarrow \ell: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} t \quad t \in \mathbb{R}$$

$$\Leftrightarrow \ell: x = 1 + 3t, \quad y = 2 - t, \quad t \in \mathbb{R}$$

b) ℓ contains the origin and is parallel to $\mathbf{b}(4, 5)$,

$$\ell: \begin{cases} x = 4s \\ y = 5s \end{cases} \quad s \in \mathbb{R}$$

c) ℓ contains the point $M(1, 7)$ and is parallel to Oy ,

$$\ell: \begin{cases} x = 1 \\ y = 7 + t \end{cases} \quad t \in \mathbb{R}$$

$\Downarrow$

ℓ has the same direction vectors as Oy
 The basis of $\mathbb{R}^2$ is (i, j)
 so j is a direction vector for ℓ
 The components of j are $(0, 1)$

d) ℓ contains the points $M(2, 4)$ and $N(2, -5)$.

$$\ell: \begin{cases} x = 2 \\ y = 4 - 9s \end{cases} \quad s \in \mathbb{R} \quad \hookrightarrow \quad \overrightarrow{MN} (0, -9) \text{ is a direction vector for } \ell$$

3.2. For the lines ℓ in the previous exercise

a) determine a Cartesian equation for ℓ ,

b) describe all direction vectors for ℓ .

3.1.a) $\ell: \begin{cases} x = 1 + 3t \\ y = 2 - t \end{cases} \quad t \in \mathbb{R} \quad \Rightarrow \quad \left. \begin{aligned} t &= \frac{x-1}{3} \\ t &= \frac{y-2}{-1} \end{aligned} \right\} \Rightarrow \ell: \frac{x-1}{3} = \frac{y-2}{-1}$

$$\Rightarrow l: x + 3y - z = 0$$

since $a(3, -1)$ is a direction vector all other direction vectors are $\{ \lambda a = (3\lambda, -\lambda) : \lambda \in \mathbb{R} \}$

$$3.1.b) \quad l: \begin{cases} x = 4s \\ y = 5s \end{cases} \quad s \in \mathbb{R} \quad \Rightarrow \begin{cases} s = \frac{x}{4} \\ s = \frac{y}{5} \end{cases} \Rightarrow l: \frac{x}{4} = \frac{y}{5}$$

$$\Rightarrow l: 5x - 4y = 0$$

the direction vectors are $\{ \lambda \cdot b(4, 5) = (4\lambda, 5\lambda) : \lambda \in \mathbb{R} \}$

$$3.1.c) \quad l: \begin{cases} x = 1 \\ y = 7 + t \end{cases} \quad t \in \mathbb{R}$$

$$\Rightarrow l: x = 1 \quad (y \text{ is arbitrary})$$

the direction vectors are $\{ \lambda j = (0, \lambda) : \lambda \in \mathbb{R} \}$

$$3.1.d) \quad l: \begin{cases} x = 2 \\ y = 4 - 9s \end{cases} \quad s \in \mathbb{R}$$

$$\Rightarrow l: x = 2$$

since $\overrightarrow{MN} = 9 \cdot j$ l has direction vectors

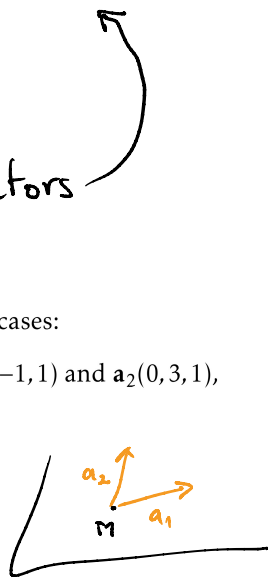
3.3. Determine parametric equations for the plane π in the following cases:

a) π contains the point $M(1, 0, 2)$ and is parallel to the vectors $a_1(3, -1, 1)$ and $a_2(0, 3, 1)$,

$$\pi: \begin{cases} x = 1 + 3t \\ y = -t + 3s \\ z = 2 + t + s \end{cases} \quad t, s \in \mathbb{R}$$

$$\Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} t + \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} s \quad t, s \in \mathbb{R}$$

$$\Rightarrow \pi: x = 1 + 3t, \quad y = -t + 3s, \quad z = 2 + t + s, \quad t, s \in \mathbb{R}$$

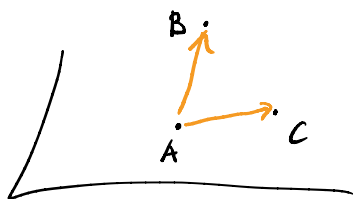


b) π contains the points $A(-2, 1, 1)$, $B(0, 2, 3)$ and $C(1, 0, -1)$,

$$\vec{AB} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \quad \vec{AC} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$$

• $\vec{AB}, \vec{AC}$ are vectors represented with points in π

• $\vec{AB}, \vec{AC}$ are linearly independent so they form a basis of the direction space $D(\pi)$



$$\pi: \begin{cases} x = -2 + 2t + 3s \\ y = 1 + t - s \\ z = 2 + 2t - 2s \end{cases} \quad t, s \in \mathbb{R}$$

$$\Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} t + \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} s \quad t, s \in \mathbb{R}$$

c) π contains the point $A(1, 2, 1)$ and is parallel to i and j ,

$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ since by convention (i, j, k) is a basis of $\mathbb{V}^3$

$$\pi: \begin{cases} x = 1 + t \\ y = 2 + s \\ z = 1 \end{cases} \quad t, s \in \mathbb{R} \quad \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad t, s \in \mathbb{R}$$

d) π contains the point $M(1, 7, 1)$ and is parallel to the coordinate plane Oyz ,

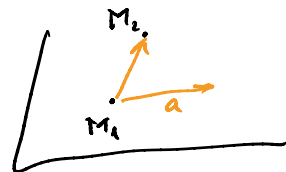
a basis for $D(Oyz)$ is (j, k) $j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\pi: \begin{cases} x = 1 \\ y = 7 + t \\ z = 1 + s \end{cases} \quad t, s \in \mathbb{R} \quad \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad t, s \in \mathbb{R}$$

e) π contains the points $M_1(5, 3, 4)$, $M_2(1, 0, 1)$ and is parallel to the vector $a(1, 3, -3)$,

$\vec{M_2M_1}(4, 3, 3)$ is a vector represented with points in π so it is parallel to π

$\vec{M_2M_1}$ and a are linearly indep. vectors
 $\Rightarrow$ they form a basis of $D(\pi)$



$$\pi: \begin{cases} x = 5 + 4t + \lambda \\ y = 3 + 3t + 3\lambda \\ z = 4 + 3t - 3\lambda \end{cases} \quad t, \lambda \in \mathbb{R} \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} + t \begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ -3 \end{bmatrix} \quad t, \lambda \in \mathbb{R}$$

f) π contains the point $A(1, 5, 7)$ and the coordinate axis Ox.

$Ox \subseteq \pi \Rightarrow O \in \pi \Rightarrow \vec{OA}(1, 5, 7)$ is parallel to π

$\Rightarrow i(1, 0, 0)$ is parallel to π

since $\vec{OA}$ and i are linearly independent, they form a basis of $D(\pi)$

$$\pi: \begin{cases} x = t + \lambda \\ y = 5t \\ z = 7t \end{cases} \quad t, \lambda \in \mathbb{R} \Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad t, \lambda \in \mathbb{R}$$

3.4. Determine Cartesian equations for the plane π in the following cases:

a) $\pi: x = 2 + 3u - 4v, y = 4 - v, z = 2 + 3u$ with $u, v \in \mathbb{R}$,

b) $\pi: x = u + v, y = u - v, z = 5 + 6u - 4v$ with $u, v \in \mathbb{R}$.

$$a) \pi: \begin{cases} x = 2 + 3u - 4v \\ y = 4 - v \\ z = 2 + 3u \end{cases} \quad u, v \in \mathbb{R} \Leftrightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} + u \cdot \underset{a}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}} - v \cdot \underset{b}{\begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}} \quad u, v \in \mathbb{R}$$

$$\pi: \begin{vmatrix} x-2 & y-4 & z-2 \\ \underset{a}{\rightarrow} 1 & 0 & 1 \\ \underset{b}{\rightarrow} 4 & 1 & 0 \end{vmatrix} = 0 \Leftrightarrow (x-2) \cdot (-1) - (y-4) \cdot (-4) + (z-2) = 0$$

$$\Rightarrow \pi: -x + 4y + z - 16 = 0$$

$$b.) \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} + u \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix} + v \begin{bmatrix} 1 \\ -1 \\ -4 \end{bmatrix} \quad u, v \in \mathbb{R}$$

$$\Rightarrow \pi: \begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = 0 \Leftrightarrow x \underbrace{\begin{vmatrix} 1 & 6 \\ -1 & -4 \end{vmatrix}}_2 - y \underbrace{\begin{vmatrix} 1 & 6 \\ 1 & -4 \end{vmatrix}}_{-10} + (z-5) \underbrace{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}_{-2}$$

$$\Rightarrow \pi: x + 5y - z + 5 = 0$$

3.5. Determine parametric equations for the plane π in the following cases:

a) $\pi: 3x - 6y + z = 0,$

b) $\pi: 2x - y - z - 3 = 0.$

a.) choose x and y as parameters $\begin{cases} x = x \\ y = y \\ z = -3x + 6y \end{cases}$

$$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix} \quad t, s \in \mathbb{R}$$

we relabel x and y

b.) choose $x = t$ and $z = s$ as parameters, then $\begin{cases} x = t \\ y = 2x - z - 3 = 2t - s - 3 \\ z = s \end{cases}$

$$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

3.6. Show that the points $A(1, 0, -1)$, $B(0, 2, 3)$, $C(-2, 1, 1)$ and $D(4, 2, 3)$ are coplanar.

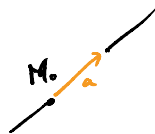
A, B, C, D coplanar $\Leftrightarrow \vec{AB}, \vec{AC}, \vec{AD}$ linearly dependent, true since

$$\vec{AB} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} \quad \vec{AC} = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \quad \vec{AD} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \quad \text{and} \quad \begin{vmatrix} -1 & -3 & 3 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{vmatrix} = -4 + 12 - 24 - 12 + 24 + 4 = 0$$

3.7. Determine parametric equations for the line ℓ in the following cases:

a) ℓ contains the point $M_0(2, 0, 3)$ and is parallel to the vector $a(3, -2, -2)$,

$$\ell: \begin{cases} x = 2 + 3t \\ y = -2t \\ z = 3 - 2t \end{cases} \quad t \in \mathbb{R} \Leftrightarrow \ell: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix}$$



b) ℓ contains the point $A(1, 2, 3)$ and is parallel to the Oz -axis,

$\Rightarrow k$ is a direction vector, $k = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$\ell: \begin{cases} x=1 \\ y=2 \\ z=3+t \end{cases} \quad t \in \mathbb{R} \Leftrightarrow \ell: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

3.8. Give Cartesian equations for the lines ℓ in the previous exercise.

$$3.7.a) \ell: \begin{cases} x=2+3t \Rightarrow t=\frac{x-2}{3} \\ y=-2t \Rightarrow t=\frac{y}{-2} \\ z=3-2t \Rightarrow t=\frac{z-3}{-2} \end{cases} \Rightarrow \frac{x-2}{3} = \frac{y}{-2} = \frac{z-3}{-2}$$

so we can choose the following two equations

$$\ell: \begin{cases} \frac{x-2}{3} = \frac{y}{-2} \\ \frac{y}{-2} = \frac{z-3}{-2} \end{cases} \Leftrightarrow \ell: \begin{cases} 2x+3y-4=0 \\ 2y-2z+6=0 \end{cases}$$

$$3.7.b) \ell: \begin{cases} x=1 \\ y=2 \\ z=3+t \end{cases}$$

Notice that z is arbitrary

$$\text{so } \ell: \begin{cases} x=1 \\ y=2 \end{cases} \Leftrightarrow \ell: \begin{cases} x-1=0 \\ y-2=0 \end{cases}$$

3.9. Determine parametric equations for the line contained in the planes $x+y+2z-3=0$ and $x-y+z-1=0$.

$$\ell: \begin{cases} x+y+2z-3=0 \\ x-y+z-1=0 \end{cases} \quad (*)$$

$$\begin{pmatrix} 1 & 1 & 2 & -3 \\ 1 & -1 & 1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 2 & -3 \\ 0 & -2 & -1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 2 & -3 \\ 0 & 1 & 1/2 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 3/2 & -2 \\ 0 & 1 & 1/2 & -1 \end{pmatrix}$$

so system (*) is equivalent to

$$\begin{cases} x + \frac{3z}{2} - 2 = 0 \\ y + \frac{z}{2} - 1 = 0 \end{cases}$$

if we choose the parameter $t = \frac{2}{2}$ then

$$\begin{cases} x = 2 - 3t \\ y = 1 - t \end{cases} \quad \Rightarrow \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ -1 \\ 2 \end{pmatrix}$$

3.10. In each of the following, find a Cartesian equation for the plane containing the point Q and the line ℓ .

a) $Q = (3, 3, 1)$, $\ell: x = 2 + 3t, y = 5 + t, z = 1 + 7t$ with $t \in \mathbb{R}$,

b) $Q = (2, 1, 0)$, $\ell: x - y + 1 = 0, 3x + 5z - 7 = 0$.

a) $Q = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix}$ and $\ell: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ 7 \end{pmatrix} \quad t \in \mathbb{R}$

$\parallel$ $\parallel$
A v

so the plane contains the points Q and A
[is parallel to v
proceed as for Exercise 3.3.e)

b) $\ell: \begin{cases} x - y + 1 = 0 \\ 3x + 5z - 7 = 0 \end{cases}$

transform these equations into parametric equations for ℓ
then, proceed as for point a.)